

AI-DRIVEN SOFTWARE TESTING & QUALITY ASSURANCE

The Software Testing & Quality Assurance track builds quality engineers and SDETs for modern, AI-driven testing. Trainees master the full testing lifecycle — test design, automation, defect management — across functional, mobile, performance, security, API, and database testing, with programming (Java, JavaScript, .NET) for automation and ISTQB certifications (Foundation, Agile, Performance, Mobile, GenAI, AI). They also apply Generative AI to testing. Graduates work as SDETs, quality, performance, mobile, security, GenAI-testing, and API testing engineers.

JOB PROFILES

SDET / Test Automation Engineer ·
Quality Engineer (QA & Process Improvement) · Performance Testing Engineer ·
Mobile Test Automation Engineer · Security Testing Engineer ·
AI-Augmented / GenAI Testing Engineer · API Test Automation Engineer



TOTAL HOURS

1,362hrs

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



CERTIFICATIONS

8

Industry-certified



Where graduates work

VOIS

Vodafone Egypt

Noon

Etisalat Misr

Expleo

Fawry

Suz Canal Bank

Orange

DXC

Aman

National Bank of Egypt

Bank Misr

TechLabs London

EBC Bank



Certifications earned

- ISTQB-CTFL** — Certified SW Testing Foundation Level
- ISTQB-GenAI** — Certified Gen-AI SW Tester
- ISTQB-AI** — Certified AI SW Tester
- ISTQB-Performance** — Certified Performance SW Tester
- ISTQB-Agile** — Certified Agile SW Tester
- ISTQB-Mobile** — Certified Mobile SW Tester
- ISTQB-Security** — Certified Security SW Tester
- ISTQB-Automation** — Certified Test Automation Engineer



Selection process

01

IQ & English Exam



Online exam covering problem-solving aptitude and English proficiency.

02

Technical Exam



Computer-based assessment covering the program's core technical fundamentals.

03

Technical Interview



Interview with the technical committee.

04

Interpersonal Interview



Candidates who pass the technical interview are invited to this final stage.



Who may apply

Open to graduates in Computer Engineering ,Computer Science and Any Background with a minimum grade of Good or higher.

Program content – 1,362 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● Fundamental Courses	267 hrs	20%	Operating Systems Fundamentals, Database Fundamentals, Computer Networks Fundamentals, Introduction to Programming, Object-Oriented Programming Concepts, Agile Software Development Methodologies, Linux Operating System, HTML5 & CSS3, Problem Solving, Java Programming, Introduction to Gen AI & Prompt Engineering
● Core Courses	738 hrs	54%	XML Fundamentals, Introduction to Generative AI and Prompt Engineering, ISTQB Foundation Software Testing Certification, Developing & Testing Database using Sql Server, Software Configuration management, ISTQB Mobile Applications Testing, Software Project Management, Effective Test case and bug report writing techniques, ISTQB CT-GenAI, ISTQB CT-AI, LLM & RAG Testing, AI Model Monitoring, Drift Detection & LLM Output Evaluation, API Development & Testing, Web Automation with Playwright, Cypress for E2E Testing, AI-Assisted Test Automation Authoring, Developing & Testing JavaScript, Developing & Testing .Net Applications, Software Quality assurance essentials, Web Performance Testing, Security testing, Mobile Applications Testing, Accessibility Testing (WCAG), Test Data Management & Synthetic Data Privacy, ISTQB Foundation Agile Tester Extension Certification, Testing Analysis and Automation, Testing Workshops, Effective Test case and bug report writing techniques, ISTQB Performance Testing
● Soft Skills	57 hrs	4%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor
● Graduation Project	300 hrs	22%	

Hands-on labs are conducted in-person. Online components cover theory and self-study modules on MaharaTech.

Student Deliverable

Every graduate must obtain an international certification relevant to their specialisation – ISTQB certifications (Foundation, Agile, Performance, Mobile, GenAI, AI, Security, Automation), confirming professional readiness for the field.

Before you apply

Explore these resources before your interview. Applicants are expected to have reviewed this material.

- Introduction to software testing
<https://maharatech.gov.eg/course/view.php?id=2>
- ISTQB Foundation Level V4.0 – From Basics to Certification
<https://maharatech.gov.eg/course/view.php?id=2372>
- Testcase Writing And Bug reporting
<https://maharatech.gov.eg/course/view.php?id=4>
- C Programming From Basics to Mastery
<https://maharatech.gov.eg/course/view.php?id=2247>
- Mastering Object-Oriented Programming (OOP) using C++
<https://maharatech.gov.eg/course/view.php?id=2206>
- ITI Values & Mission
<https://iti.gov.eg/about-us>

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.